

NAVCORE-SoOP

Order to first flight

The complete runbook, with an audit of what is still manual,
soldered by hand, or not yet measured

Board	45.1 × 46.1 mm, 6-layer, STM32H743VIT6
Gate verdict	READY TO ORDER — all 43 gated prerequisites proven
Hardware verdict	NOT READY TO FLY — 9 items need hardware
Toolchain	KiCad 9.0.8, ArduPilot Copter-4.7.0, Python 3 + pcbnew

*Every figure in this document was re-derived from the repository's own tooling on 19 September 2026 rather than copied from another document. Where a number is a judgement, an assumption or genuinely unknown, it says so. Sections marked **flag** are the ones capable of costing money, a board, or a bench session.*

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1 How to use this, and how to verify it

This is one document in the order you actually do things: order the board, order the aircraft parts, assemble, power it up, and fly it. It exists because the same information was spread across docs/ORDER.md, docs/ASSEMBLY.md, docs/BUILD.md and docs/BUYING.md, and those four had drifted apart — in three cases far enough apart to send you shopping for parts with nowhere to put them.

Verify before you spend. Read the verdict, not a summary:

```
python3 tools/preflight.py → must print READY TO ORDER
```

The verdict is *derived* from a manifest of 43 gated prerequisites, so an unrun tool, a non-zero exit, a failed check or an unclassified manifest entry is a blocking failure. It cannot be typed by hand and it cannot shrink silently. What it proves is that the *design* is orderable. It cannot prove anything about money, stock, or JLCPCB's own upload preview — those are Sections 2.8 onward.

1.1 The four commands that matter

python3 tools/preflight.py	the gate; must say READY TO ORDER
bash tools/make_order_bundle.sh	regenerate fab/ and zip the order
bash tools/finish.sh	the whole design pipeline, ending in the fab outputs
bash tools/build_firmware.sh	build arducopter.apj, pinned to Copter-4.7.0

1.2 Freshness

fab/stock-snapshot.json is re-fetched from jlcpcb.com and tools/check_stock.py blocks the gate once it is older than **7 days**. It was last refreshed 2026-09-19 and resolves **56 of 56** non-DNP codes with cover for two boards. Everything else in fab/ is regenerated by tools/make_order_bundle.sh, which refuses to build the bundle if the copper layer count is not 6.

2 Part 1 — Order the bare board and its assembly

2.1 Regenerate, do not trust

```
bash tools/make_order_bundle.sh
```

This re-exports the gerbers and drill, re-runs gen_bom.py for *all three* variant pairs, checks the layer count, and writes fab/NAVCORE-SoOP-order-bundle.zip.

Do not hand-copy or hand-edit anything in fab/. A cp of a stale variant into the bundle is exactly how the Economic BOM once shipped quoting Y2's superseded, unbuyable part code for three days — and fab/ORDER.md was recommending those files at the checkout while it did.

2.2 What to upload

File	Where
fab/gerbers/ — zip the <i>contents</i>	PCB gerbers + Excellon drill
fab/BOM-NAVCORE-SoOP.csv	assembly BOM, the default build
fab/CPL-NAVCORE-SoOP.csv	pick-and-place, the default build

Or upload the bundle zip, which contains those plus HOW-TO-ORDER.txt. fab/gerbers/ holds **16 files**: 14 .gbr (6 copper, 2 paste, 2 silkscreen, 2 mask, edge cuts, drill map), NAVCORE-SoOP.dr1, and NAVCORE-SoOP-job.gbrjob.

Stop-condition at upload. If the preview shows **four** copper layers, abort. The power planes are missing and the board will not work. It must show six.

2.3 Fabrication settings

Setting	Value	Why
Layers	6	In1 is a solid GND plane; In4 carries the rails
Dimensions	45.1 × 46.1 mm	
Thickness	1.6 mm	30.5 mm stack standard
Outer copper	1 oz	every current figure assumes it
Surface finish	ENIG	0402s and LGA-14s want a flat finish
Min track / spacing	0.1016 mm (4 mil)	should be free tier
Min via / drill	0.45 mm / 0.20 mm	
Impedance control	No	USB is a plain differential pair
Stencil	not needed	everything machine-placed but Section 7

The 1 oz line is not cosmetic: **every current-capacity figure in `check_build.py` assumes it**, and 0.5 oz halves each trace’s rating.

2.4 Quantity, and the tier decision

Order 5 bare boards and 2 assembled. Five is the multilayer minimum; two is the usual SMT assembly minimum.

	Standard (panelised)	Economic
Board minimum	70 × 70 mm , so 2 × 2 panel + rails	10 × 10 mm, no rails
Places	everything, including U3	everything except U3
Feeder fees	included	GBP 40–46 still charged
Hand-fitting	none	U3 (ICM-42605, second IMU)

Recommendation: Standard, panelised 2 × 2, “panel by JLCPCB”. Do not commit a panelised board file — the board files stay exactly as gated, so nothing has to go through DRC, connectivity and the whole gate again. A hand-panelised `.kicad_pcb` would have to, for no benefit.

If you take Economic instead, **take its BOM and CPL as a pair** (`*-economic.csv` for both). Mixing an Economic BOM with the default CPL hands JLC a placement list for parts the BOM does not order. `tools/check_variants.py` — now a gated check — fails if any pair disagrees with the board.

2.5 The variant files, and why two of them are identical

Variant	BOM lines	Placements	Difference from default
NAVCORE-SoOP (default)	68	129	—
NAVCORE-SoOP-economic	68	128	omits U3 only
NAVCORE-SoOP-nofpv	68	129	none — byte-identical

`-nofpv` is byte-identical to the default pair because the 9 V VTX block is DNP *either way* (`design.POPULATE_VTX` is `False`). It is a label, not a saving. Choosing Economic only makes sense if you want to hand-fit the second IMU.

2.6 Parts worth checking before you pay

Stock is no longer quoted in prose. It lives in the dated snapshot `fab/stock-snapshot.json`, age-gated at 7 days by `tools/check_stock.py`.

Two substitutions are **not optional**:

- **Y1 must be C2682774** — a *passive crystal*, CL 20 pF. The visually identical C2901556 is an **active oscillator**: same SMD3225-4P land pattern, but pin 4 is VDD, which this board grounds. Substituting it gives a board that never starts.
- **Y2 must be C5563878**. Its predecessor C22381771 was down to **1 unit** with no published restock date when `check_stock.py` was written.

2.7 The DNP list — derived, never typed

14 designators ship unpopulated: C68 C69 C70 C73 L5 Q3 R15 R40 R41 R42 R43 R45 U11 U18.

Block	Why it is not fitted
The 9 V VTX buck (L5, Q3, U18, C68–C73, R41–R45)	POPULATE_VTX is <code>False</code> because its switch node could not be closed on this placement. A VTX runs from +5 V instead.
U11, the CAN transceiver	a <i>thermal</i> decision, not a CAN one: it was fitted but never enabled in <code>defaults.parm</code> . Removing it dropped U9's peak from 445 mA / 179°C to 375 mA / 157°C.

Both have pads, and the gate prints [NOT FITTED] for each on every run. Do not type this list — `make_order_bundle.sh` derives it from the BOM into `HOW-TO-ORDER.txt`.

2.8 Check these on screen at upload

- Outline and board size: **45.1 × 46.1 mm**
- Copper layer count: **6**
- Drill count and the 0.20 mm minimum
- **CPL orientation against their preview render** — the one thing no local tool substitutes for
- **1 oz outer copper**, explicitly
- **Panel by JLCPCB**. Standard assembly requires a single PCB of at least 70 × 70 mm and this board is 45.1 × 46.1 mm, so the panel and its breakaway rails have to come from somewhere. Their own panelisation is the low-risk answer: you then confirm the panel outline in the same preview as the CPL.

2.9 Paying

- **Coupons are once-per-order** and cannot be combined. Take the SMT one.
- **6-layer promotions recur**; worth waiting a few days if you are not in a hurry.
- **UK import VAT is 20%**, collected by the courier above GBP 135 with an €8–12 handling fee. It is not a JLCPCB charge and is easy to forget when comparing against a UK-stocked alternative.

3 Part 2 — Order the aircraft parts

Order these at the same time; lead times run 2–4 weeks.

Item	Qty	Note
TBS Source One V5 7in DC frame	1	320 mm wheelbase, 6.0 mm arms; open-source DXF
SpeedyBee BLS 60A 4-in-1 ESC	1	30.5 × 30.5 stack
2806.5 1300KV motors	4	19 × 19 bolt pattern, M5 shaft
Gemfan 7040 props	4 + spares	5.0 mm bore

Item	Qty	Note
Zeee 4S 6500 mAh pack	1+	138 × 47 × 48 mm, 615 g, EC5
EC5-to-XT60 adapter	2	the pack is EC5, everything else is XT60
M10 GNSS + QMC5883L compass	1	comes with a JST-GH 6-pin lead
ELRS receiver, ESP-based	1	an STM32 ELRS receiver cannot carry MAVLink
M3 × 14 socket screws	16	motors, skid sandwiched
M3 × 30 socket screws	4	the stack bolts
M3 female-female standoffs, 12 mm	4	the ESC-to-FC spacer
M3 socket assortment (10/12/14/16/20) + washers	1 kit	the anti-reorder kit
Frame kit's own 30 mm / 22 mm stand-offs	—	already in the kit; do not buy more
TPU skids, printed	4	3.5 mm thick, 40 mm drop, 19 × 19 holes
JST-SH 1.0 mm 4-pin pigtails	pack of 10	J5, then J9/J11
JST-SH 1.0 mm 8-pin pigtail	1	J2, the ESC lead. A mismatch is re-pinnable with a fine pick, but owning one removes even the rework
JST-GH 1.25 mm 6-pin spare lead	1	closes the J3 plug risk
5 V passive piezo buzzer	1	passive, not active
microSD card	1	32 GB
Radxa Zero 3W companion	1	carries the flow camera. The <i>tray</i> is a 3D print; the board is a purchase
OV9281 global-shutter module	1	the flow camera, mounted on the companion
XIAO ESP32S3 Sense	1	the nose recording camera, writes to its own microSD
M3 silicone grommet, spare set	1 pack	the flange width belongs to the frame kit's grommets, so a wide one used to mean a different grommet. A spare set makes it a swap, not an order
Bench PSU with current limit	1	Section 8 does not work without one
Multimeter with continuity	1	for buzzing the ESC cable
Smoke stopper (XT60)	1	first battery connection only
Thermocouple or IR thermometer	0–1	only for U8 now; U19 logs U9
ST-Link V2 + fine wires or pogo pins	1	for the SWD bootloader
Nooelec SAWbird+ IR (1620 MHz)	1	own this by T3b, not later
Nooelec 1620 MHz patch antenna	1	own this by T3b
RTL-SDR with a bias tee (Blog V3/V4)	1	not a generic R820T2 stick. The SAW-bird+ is powered <i>up the coax</i> at 3.3–5.5 V, which only a bias-tee dongle supplies — the Blog V3/V4 gives 4.5 V at up to 180 mA in software. On a generic stick the LNA sits unpowered and T3b cannot be started at all. An external bias tee (or Nooelec's Iridium bundle, which includes the patch, a DC block and the adapters) is the alternative

Do not buy a PMW3901 or a VL53L1X. Both were deleted from the design in the Rev B re-layout and have no pads. docs/BUYING.md told you to hand-fit them until 2026-09-19, and priced a “GBP 227 hand-fit route” around two parts that cannot be bought a place for.

3.1 The one order — nothing here should make you order twice

Every “measure this, then use a different part” branch in this document is closed by buying the alternatives up front. The premium is about **GBP 6**.

What to add	What it closes
M3 socket assortment + washers	Both derived joints round <i>up</i> past their engagement target (4.5 mm and 6.7 mm into the thread) and the skid’s tapped depth is the only measurement that can invalidate the length you picked.
4 × 12 mm female-female spacers	the ESC-to-FC separation is 12.1 mm and <code>fasteners.py</code> says plainly that a grommet cannot hold it. Without these the stack cannot be assembled at all.
the kit’s own standoffs	do <i>not</i> buy 35 mm standoffs for the top plate; it has no 30.5 mm holes.
spare JST-GH 6-pin lead	the residual J3 risk is a <i>bulky moulded boot</i> on the lead you receive, not the board: the 5.30 mm clear run meets its derived 3.6 mm requirement. Test-fit the plug before soldering; a second lead is then the fallback rather than an interruption.
10 JST-SH pigtails, not 4 SAWbird+ and a bias-tee RTL-SDR with Tier 1	the first one you crimp badly is then not a project stall. T3b cannot be started without them, and the board does not supply them. A generic R820T2 stick has no bias tee , so the SAWbird+ would sit unpowered — this is the one place a cheap substitution silently costs you the whole measurement.
Radxa Zero 3W, OV9281, XIAO ESP32S3	the companion, the flow camera and the recording camera are <i>purchases</i> , not prints, and the two flow bench items cannot be closed without the first two. Buy them on this order and only the 3D-printed tray is left outstanding.

Three things need the parts in hand, and **only two of them are 3D prints**: the camera mount and the Radxa Zero 3W’s tray (the vendor never published its hole pattern). The third is the frame kit’s top-plate standoff spacing, which the kit supplies. Read that carefully: the *prints* are deferred, but the Radxa board and both cameras are **purchases**, and they are on the list above on purpose — buy them in this order and the only thing left outstanding is something you can print.

3.2 The bench measurements that could have cost you a purchase

Sort the bench items into two kinds, because only one kind can reach your wallet. Six of them cannot: BATT_AMP_PERVLT, the IMU orientation, FLOW_ORIENT_YAW, flow quality over grass, and both thermal readings are numbers you write down or calibrate, and no possible answer changes what you should have bought. U8’s junction is worse than those — if it really is 139°C the remedy is a **respin** — but it is still not a purchase.

Three could have. Each is now bought around:

Measurement	Why it can no longer send you shopping
J2's ESC cable pinout	a wrong cable is re-pinnable with a fine pick <i>and</i> a spare 8-pin JST-SH lead is on the order — previously the only route was to rework the cable you already had
the RF chain can hear Iridium	the SAWbird+ is powered <i>up the coax</i> , and the order now names a bias-tee dongle (Blog V3/V4) rather than "an RTL-SDR". A generic R820T2 stick would have turned a small saving into an unpowered LNA and a stalled T3b
the grommet flange	the flange is a property of the frame kit's grommets, so a wide one meant a different grommet. A spare silicone set is now a line item: measure, and swap rather than order

The honest residue: one answer can still cost real money, and it is not a part. If U8 is genuinely at 139°C the remedy is a new board — which is why **three spare bare PCBs** are on the order and why T3a is a stop gate rather than a note. `tools/fasteners.py` prints this list itself under “*not derived — these need the parts in hand*”.

4 Part 3 — While you wait (free, and the highest-value work)

1. **Register ArduPilot board ID 7180.** Not needed to order; needed before sharing. The request is written out in `firmware/BOARD-ID-REQUEST.md`, with the registry check, the alternates and the issue text. **Do not request 9001** — it is free but it is above the #7199 ceiling that `board_types.txt` explicitly asks you not to cross, while 4544 free IDs sit below it.
2. **Read `docs/ASSEMBLY.md` Stages 3 and 4 twice.** The two screw-depth warnings are the ones that strip a frame.
3. **Print `docs/img/padmap.svg` and the F.Silkscreen gerber 1:1 on paper.** Nine pads arrive with no label (Section 14).
4. **Decide the USB edge orientation before cutting any loom.** All four edges carry a connector; there may be exactly one orientation that works.
5. **Dry-run the SITL suite** if you have a Linux box: `bash sitl/run_scenarios.sh`.

5 Part 4 — On arrival: inspect before you spend more

5.1 Visual inspection

Under magnification, look at U1 (**LQFP-100, 0.50 mm pitch**), U2/U3 (**LGA-14**, no accessible leads), and U13 (**TQFN-28 with an exposed pad**). You are looking for bridges and tombstones. JLC is good at this. Look anyway.

5.2 Confirm the machine-readable facts

```
python3 tools/preflight.py    never mind the board — confirm DRC, ERC, gerbers and
                             BOM/CPL are all current
python3 tools/check_variants.py every variant pair matches the board
```

5.3 The frame measurements that settle a real decision

Check with calipers	Parsed value
top plate width	42.50 mm — the battery is 47 mm and already overhangs 2.25 mm each side; there is no room for an upward ToF beside it
bottom plate M3 positions	$x = \pm 14.60$ and ± 11.00 , no 30.5 mm pattern — so the FC cannot bolt under the bottom plate; that idea is closed
which plate is top	inferred from length (160.26 mm). The one genuinely open row

5.4 The three things no drawing can answer

- **microSD withdrawal clearance** — depends how *you* mount it
- **grommet compression under load** — a property of the rubber
- **the real TPU gear's shape, hole pattern and strength** — the assembly is modelled as four legs at the motor pattern on assumed dimensions

6 Part 5 — Mechanical assembly

6.1 Stack height

22.3 mm against 25 mm of inner space — 2.7 mm spare. Two figures exist and both are correct about different things: `check_build.py` gives **19.8 mm** for bare parts, and `check_mechanical.py` gives **22.3 mm** including J3's mated plug. **22.3 mm is the one the top plate must clear.**

6.2 The two screw warnings — read both

```
python3 tools/fasteners.py
```

- **Motor to arm, skid sandwiched.** Needs 13.5 mm, so order M3 $\times$ 14. But rounding 13.5 $\rightarrow$ 14 drives **5.5 mm into a thread designed for 4.0 mm**. A motor boss is often only about 4 mm deep, and **a bottomed-out screw feels tight while clamping nothing** — that is the failure that throws a motor in flight. Measure the tapped depth before fitting sixteen screws. M3 $\times$ 12 plus a washer is the usual fix.
- **FC to ESC to frame, the 30.5 mm stack bolt.** Needs **27.3 mm**, so M3 $\times$ 30. Rounding 27.3 $\rightarrow$ 30 puts **6.7 mm** into the press nut. If the tapped depth is shallower than that the screw bottoms out and clamps nothing.

This is the whole reason the buy list carries an M3 assortment rather than one length. `free_island` is not involved; the point is that both joints are decided by a measurement you have not taken yet, so own every plausible length.

6.3 Screw heads against copper — 0.08 mm

No copper is under any screw head, and the margin is thin. The closest is TP20 at **2.83 mm** against a **2.75 mm** cap-head radius — **0.08 mm**, and that assumes **no washers**. A washer, or a head even 0.1 mm over nominal diameter, puts metal on copper. If you add washers, re-measure before trusting this line.

6.4 Sequence

1. **Inspect the boards** (Section 4.1).
2. **Print the fab drawing and silkscreen 1:1.** Nine unlabelled pads — Section 14.
3. **Dry-fit the FC into the centre plate.** Confirm nothing blocks the bottom edge where the microSD ejects.
4. **Buzz the ESC cable** both ends against the J2 pinout below, before anything is powered.
5. **Sit a skid under each arm,** holes aligned. Motor on top, wires inboard. Thread all four screws by hand before tightening any; tighten diagonally.
6. **The stack, bottom to top: ESC, grommets, FC.** Fit the silicone grommets into the FC's four mounting holes; ESC into the frame on its 30.5 mm pattern; FC on top with USB-C facing the direction you will actually plug from.
7. **4 × M3 × 30 through the stack,** with the four 12 mm female-female spacers setting the ESC-to-FC separation.
8. **30 mm standoffs for the top plate** — from the frame kit.

6.5 J2, the ESC connector — confirmed rather than assumed

Wired **1 GND, 2 VBAT, 3 M1, 4 M2, 5 M3, 6 M4, 7 CUR, 8 TEL** (pins 9/10 are the shell tabs). Betaflight's board page for the SpeedyBee F405 V4 gives the same FC-side order, so it **matches** — no cable rework needed.

The vendor table reads oddly because it lists the two *ends* of the cable in opposite order, since they face each other. Reverse the ESC column and every pair lines up: GND↔GND, BAT↔VBAT, M1↔S1, ..., CUR↔CURRENT.

Buzz it anyway. One automated read of a wiki table is not the same as measuring the cable you received. An 8-pin JST-SH is re-pinnable with a fine pick — a mismatch costs a cable rework, not a respin.

Pin 8 is inert. The ESC end is N/A — stock BLHeli_S has no telemetry output at all. SERIAL5_PROTOCOL 16 and the UART8_RX wire are correct and cost nothing, but they will receive nothing until a telemetry-capable ESC (BLHeli_32, AM32) is fitted. That is why the board ships INS_HNTCH_MODE 1 (throttle) rather than 3 (RPM).

7 Part 6 — Soldering: four joints minimum to fly

Everything else plugs in.

- **Buzzer** — PZ1 (+) and PZ2 (−). Must be a **passive** piezo: PA15 is a timer channel and ArduPilot generates the tone patterns itself. An active buzzer has its own oscillator and loses every arming and failsafe pattern.
- **LED strip, optional** — PL1 = data, PL2 = +5 V, PL3 = GND. **PL3 has no label.** 5 V WS2812B.
- **Do not solder to P41.** It is **5 V** and unlabelled. Identify it on the printed gerber first, never by counting from a neighbour.

8 Part 7 — T1: first power, before the MCU sees anything

USB disconnected. No battery. Do not skip this. A board that goes straight to USB can destroy the MCU on a rail fault that a current-limited supply would have caught for nothing.

Bench supply on VBAT (J2.2), **16.8 V, current limit 100 mA.**

#	Check	Expect	If wrong
1.1	current at power-on	< 80 mA	something is shorted: kill it and inspect U8, U18 and the bulk caps
1.2	+5V rail	4.97 V ± 0.2	U8 TPS54202, L1, feedback divider
1.3	+3V3 rail	3.3 V ± 0.1	U9 AP2112
1.4	+3V3A rail	3.3 V ± 0.1	U10 TLV75533
1.5	+9V rail	not populated	U18 is DNP by design

Raise the limit to 500 mA only once every rail is correct.

If a rail is dead with a correct supply, look at R46 before the buck. It is Q4's gate pull-down. Without it the gate floats, the FET never turns on, and the board is dead with everything wired correctly.

Reverse polarity is protected, but do not test it. Q4 (WST4041 P-FET) blocks a reversed pack and DZ1 clamps the gate, but D1 is still a TVS rather than a series diode. Use the XT60 keying and check twice.

USB cannot power this board. VBUS runs the USB PHY only; the rails come from VBAT. That would need a respin. For bench work, supply VBAT even to configure it.

9 Part 8 — T2: MCU alive

- Bootloader over SWD first, before USB does anything.** Flash the board's own bootloader from Tools/bootloaders/ with dfu-util, using TP20 (GND) and TP21 (+3V3) as the probe's reference. A board with no bootloader will not enumerate.
- Data-capable USB-C cable — charge-only leads enumerate nothing. Keep the supply connected.
- Hold BOOT, tap RESET. lsusb should show 0483:df11.
- Flash arducopter.apj from tools/build_firmware.sh, pinned to Copter-4.7.0.
- Connect a ground station; confirm the board ID and read **every** boot message — every fitted sensor should probe, and nothing unfitted should be claimed.
- Copy copter-deadreckon-home.lua to the microSD at APM/scripts/ by hand.** The build script does not package it into the .apj. Without it, FS_EKF_ACTION 0 ships with no GPS-loss response at all.

10 Part 9 — T3: sensors on the bench, props OFF

#	Check	How	Note
3.1	IMU1 (U2) orientation	roll/pitch, watch the horizon	asserted in hwdef, never measured
3.2	IMU2 (U3) orientation	same, INS_USE2	only if U3 was fitted
3.3	Barometer	tracks a 1 m lift	
3.4	Battery voltage	against a meter	BATT_VOLT_MULT 11.000, should be close
3.5	Battery current	known load	ships BATT_AMP_PERVLT 25.0 — a starting point, not a calibration

#	Check	How	Note
3.6	RC input	RC_PROTOCOLS 512 (CRSF)	set to 1 for SBUS/FPort
3.7	ESC telemetry	RPM with SERIAL5_PROTOCOL 16	receive-only, and inert on a BL-Heli_S ESC
3.8	Motor order and direction	motor test, props off	J2.3–J2.6 = M1–M4
3.9	Buzzer, LEDs	arming tone, strip lights	
3.10	TFS20-L, if fitted	tape measure at 1 / 5 / 10 m	RNGFND1_TYPE 46

Everything optional ships disabled in `defaults.parm` and must be turned on here. That is deliberate: a rangefinder configured but absent fails prearm, and a bare board arms out of the box. Do not enable a sensor you have not fitted.

11 Part 10 — T3a: thermal, and the board measures half of it for you

U19 (TMP119) measures U9 for you. It sits 3.9 mm from U9 on I2C1 at 0x48, and `defaults.parm` ships `TEMP_LOG 1`, so every flight writes a `TEMP` message to the card.

Part	Why it matters	Status
U9 (3.3 V LDO)	linear regulator dropping 1.7 V in SOT-23-5 with no thermal pad, carrying the MCU, flash and microSD	junction 104–157°C against a 150°C limit. It has thermal shutdown — being wrong means the 3.3 V rail switching off in flight, not smoke
U8 (+5 V buck)	16.8 V → 5 V at 0.95 A through a SOT-23-6 with no thermal pad, sandwiched between a 60 A ESC and a battery	junction 70–139°C against a 125°C recommended max — an opinion until a thermocouple says so

Both brackets exist because the junction-to-ambient figure is a property of *your* assembled board and the two ends come from different sources (JEDEC and EVM columns of the datasheet).

Half of that bracket is now measured rather than assumed. A SOT-23 has no thermal pad, so every watt leaves through the leads into whatever copper is attached to them — and `tools/thermal_vias.py` reads that copper off the board for *both* regulators. Measured 2026-09-19: U8.1 has **3 vias within 1.7 mm holding 7655.8 mm² of GND plane**, U9.2 has 2 vias onto the same plane, and **no legal position exists for another via on either part** (the search runs to 1.7 mm and refuses in-pad vias, which would wick solder). That is the strongest statement the layout alone can make, and it is why the low end of the bracket above is the plausible one for this board. **It still does not license a number** — copper moves the bound monotonically, and only a thermocouple converts it into degrees.

What to do: power it stack-assembled, logging to the card, ten minutes at steady state. Read `TEMP[0]`. If you own a probe, put it on U8 too and on U9's case once — U19 is 17.2 mm from U8 and cannot speak for it, and one case reading establishes the offset that U19 then carries forward alone.

Stop and ask if `TEMP[0]` exceeds about 85°C or U8's case exceeds about 100°C.

12 Part 11 — T3b: the RF chain, and the single open question

This is the one place the project is honest about not knowing.

The Doppler solve is written and validated. `tools/soop_solver.py` runs SOLVER OK, recovering a known position from clean geometry at **160.1 m median at 5 Hz**, and reports a singular geometry rather than inventing a number.

What is not known is what the shipping antenna and receiver actually deliver. The shipped `DopplerErrorModel` sigma is **180 m**; the measured divergence boundary was taken at **20 m**. The SITL scenarios print this warning beside their own results:

“The 120 m divergence boundary DOES NOT APPLY here: it was measured at sigma 20 m and this run is at sigma 180 m. Re-measure the boundary at the current sigma before reading a hold/runaway verdict off it.”

Rescaling it by arithmetic would invent a measurement. Only a bench RF measurement moves this: a reference receiver, the real antenna, real geometry. Until then, do not read a position-hold verdict off those scenarios.

13 Part 12 — T3c: IMU temperature calibration

Free, and it attacks the dominant error term. Do it once the board is warm enough to be realistic — **4S, stack assembled**, not the board in free air, because the assembly *is* the thermal environment.

14 Part 13 — T3d: the nine unlabelled pads

Of the **30** pad and test-point references that exist on the board, **21** get a silkscreen label. These **9** have no clean position at the board’s own 0.80 mm minimum text height, so `tools/silk_labels.py` leaves them **blank rather than printing something illegible** — a wrong label on a board is worse than a blank pad, because it will be trusted:

P41 P44 P62 P72 PL3 TP4 TP6 TP7 TP9

P41 is 5 V. TP6 is USART1_RX. Never identify a gold square by counting from its neighbour — use the printed gerber and `docs/img/padmap.svg`. The gate reprints this list on every run, so treat any other number you read as stale.

15 Part 14 — T4/T5: tethered, then flight, strictly in this order

1. **Run the SITL scenarios first:** `bash sitl/run_scenarios.sh`. It found three prearm faults no build check saw.
2. **Battery via the smoke stopper and the EC5-to-XT60 adapter** the first time.
3. **Tethered hover in Stabilize.** Land. Pull the log.
4. Tune `INS_HNTCH_FREQ` and `INS_HNTCH_BW` from the log’s FFT.
5. **GPS Loiter.** This is a working aircraft.
6. **Props on last**, after motor direction is confirmed.

Everything past GPS Loiter is research. GNSS-denied position hold is uncharacterised until T3b is measured.

16 Part 15 — Audit: what is still manual, soldered, or unmeasured

This is the audit you asked for. It is split into the three kinds of risk, because each is removed by a different action: *manual work* by buying a jig or an alternative, *hand soldering* by a stencil or an assembly tier, and *unmeasured values* only by a bench session.

16.1 15.1 Manual soldering

The board is machine-assembled, so the soldering is deliberately minimal. What remains:

Item	Joints	Risk and mitigation
Buzzer, PZ1/PZ2	2	low. Must be a passive piezo; fitting an active one costs you every arming and failsafe tone pattern, which you will only notice in the air
LED strip, PL1/PL2/PL3	3	low, and optional. PL3 is unlabelled — see Section 14
U3, second IMU	14-pin LGA	only if you chose Economic. An LGA-14 has no accessible leads, so it needs hot air or a stencil, not an iron. Standard assembly fits it and removes this entirely

The inefficiency, stated plainly: choosing Economic to save one part moves an LGA-14 onto your bench and does *not* save the feeder fees. It is the single manual soldering task on this board that needs real equipment, and it buys you one optional IMU. Unless you want the redundancy, Standard is the cheaper answer once your time is counted.

There is **no stencil requirement** and no hand-placed passives. That is by design.

16.2 15.2 Manual assembly

Task	Why it is manual, and what would remove it
The stack	four bolts, four spacers, grommets, the ESC and the frame's press nuts. Not automatable and not worth automating
Motor screws	16 screws whose length depends on a tapped depth you must measure. The M3 assortment removes the <i>reorder</i> , not the measurement
Skids	printed, not bought. This is a feature: the 3.5 mm thickness is yours to set, which is why the screw length is not a fixed number
Board connectors	the GPS, RC, I2C, lidar and ESC leads all plug in; no crimping except the spare pigtails
Bootloader over SWD	four wires or pogo pins to a bare board. There is no factory-installed bootloader, so this cannot be skipped
The GPS-loss applet	<code>copter-deadreckon-home.lua</code> must be copied to the microSD by hand; the build does not package it into the <code>.apj</code>
Companion mount	no mount exists yet. Deferred with the companion, so nothing blocks

Two structural constraints worth naming, because neither is removable now. Note that the first reads like a defect and is not — it is the opposite, and worth reading for how it was settled:

- J3's GPS clearance **passes, because the requirement was re-derived rather than the board changed.** The clear run before R21 is **5.30 mm**. The table's old figure of 6.0 mm was a JST-GH *class* nominal, and it assumed a bend allowance that nothing in that corridor needs: a pre-wired GH shell exits its wires at the rear, already parallel to the board, and the only neighbour is a 0402 resistor 0.55 mm tall. Shell half-width 2.6 mm plus 1.0 mm of handling margin is 3.6 mm, so 5.30 mm clears by 1.7 mm. **Relocating R21 was mapped and rejected** — TP20's courtyard and a via fence leave no position in the span that yields 6.0 mm, so there was never a fix available there. The residual risk is a bulky moulded boot, which is why a spare JST-GH lead is on the order list and T2 fits the plug *before* the headers are soldered.

- **The board is 45.1×46.1 mm, below JLCPCB's 70×70 mm Standard minimum.** Confirmed against their capability page: Standard single-PCB assembly runs 70×70 mm to 460×500 mm, while Economic runs 10×10 mm up. So Standard needs the board panelled with breakaway rails — an extra fabrication step and an extra thing to confirm in their preview, not a blocker. Economic avoids the panel entirely but charges the feeder fees anyway and drops an LGA onto your bench (Section 15.1), which is why Standard plus panelisation is the recommendation.

16.3 15.3 Measurements that have not been confirmed

This is the longest list, and every row is a value some part of the build *assumes*. The gate's own verdict names nine of them as NOT READY TO FLY.

Unconfirmed value	What it decides	Confirmed by
Motor boss tapped depth	M3 $\times$ 14 vs 12 + washer. Too deep a screw clamps nothing while feeling tight	your calipers, before the first of 16 screws. Covered by the assortment
Press-nut tapped depth (30.5 mm joint)	M3 $\times$ 30 vs 28 + washer, on 6.7 mm of engagement	your calipers
Skid thickness, assumed 3.5 mm	sets the motor screw length entirely	you, because you print them. Measure the print
Grommet flange $\leq$ 5.9 mm	at 6.0 mm it touches a printed label; a 7 mm washer overlaps by 0.52 mm	calipers on the grommets you receive
ESC cable continuity	whether J2 matches the ESC you were sent. Wrong here means a dead board or a smoked one	a multimeter, both ends, before any VBAT
J3 plug against 5.30 mm	whether the received GPS lead seats	fit the plug before the headers are soldered
BATT_AMP_PERVLT	battery failsafe and consumed-mAh. Ships 25.0; it is a property of the ESC's shunt, not of this board	a known load at T3.5
IMU orientations	ROTATION_ROLL_180 is asserted in the hwdef, never measured. A mirrored axis means it does not fly	roll and pitch it, watch the horizon (T3.1)
U8 junction temperature	70–139°C modelled against a 125°C recommended max. If it really is at 139 the fix is a respin , not a reorder	a thermocouple, T3a
U9's board-to-junction offset	makes U19's logged value meaningful rather than a nearby number	one probe reading on the case, then U19 carries it
FLOW_ORIENT_YAW	the sign of the flow correction, if flow is ever fitted	bench calibration
The RF sigma the antenna delivers	everything about GNSS-denied hold. Model says 180 m; the boundary was measured at 20 m	bench RF, T3b. The single open question
Whether this antenna can produce a usable Iridium fix at all	the project's premise	T3b
Flow quality over real grass	simulated flow is perfect flow	a field test

Unconfirmed value	What it decides	Confirmed by
Radxa Zero 3W hole pattern	the companion tray, if the companion is ever fitted	not published — measure the board, then 3D-print a tray
Frame top-plate standoff spacing	nothing to buy: the kit supplies them	the kit
MicroSD withdrawal clearance	how you mount it	your mount
Grommet compression under load	a property of the rubber	the assembled stack

16.4 15.4 The three worst, ranked

1. **The RF sigma (T3b).** It is the only item that can invalidate the project's premise rather than a build step, and no amount of desk work substitutes for it. Everything downstream of it — position-hold claims, the SITL scenarios' hold/runaway results — is explicitly not readable until it is measured.
2. **U8's junction temperature.** 139°C against a 125°C recommended maximum is a measure-and-stop gate, and the remedy if it is real is a board respin. The mitigation is that six layers with planes should land near the EVM end of the bracket — but that is an opinion until a probe says otherwise.
3. **The ESC cable and the tapped depths.** Cheap to check, catastrophic to skip. Both are on this list only because they are properties of parts somebody else manufactured, and both are silenced by owning the M3 assortment and a multimeter.

16.5 15.5 Design-level inefficiencies, for a future revision

These do not affect this order. They are recorded because a respin is the only place they can be fixed, and one of them (U8) may force one anyway.

- **USB cannot power the board.** VBUS runs the PHY only. Every bench session needs a VBAT supply, which is why T1 exists at all.
- **Nine pads arrive unlabelled** (Section 14) because a three-character label at the 0.80 mm minimum height covers a 2.84 mm pitch pad and its neighbours. More silkscreen real estate, or fewer test pads, would remove a bench session.
- **14 DNP designators still have pads and copper.** Correct for reversibility, but it is unused copper on two rails.
- **-nofpv is byte-identical to the default pair.** A variant that ships the same bytes as another is a place for someone to pick the wrong file.
- **The Economic variant saves one part and not the feeder fees,** which makes it a worse deal than it reads as (Section 15.1).
- **In4 rail fragmentation** is an accepted warning on this board.
- **The 1620 MHz SAW filter is not stocked at LCSC,** so the RF front end needs a second supplier and a separate board.

17 Part 16 — What only the outside world can settle

Section 16.3 is the authoritative register of every value this build *assumes*, and it is deliberately not repeated here — an earlier draft restated nine of its rows and the two lists were free to disagree. What

follows is the short list no local check and no desk work can close, because each needs a fab, a supplier, a vendor or a maintainer to answer it.

Item	Status
JLPCB's upload preview	outline, panel size, CPL orientation, stackup and drills. No local tool substitutes for looking at their render
1 oz outer copper	every current figure assumes it, and 0.5 oz halves every trace's rating. Confirm what actually ships
LCSC stock	a live external fact, age-gated at 7 days. Re-fetched 2026-09-19 : 56/56 codes resolve, covering 2 boards
The 1620 MHz SAW filter	not stocked at LCSC — a separate board from a separate supplier, and the only part of the RF front end this BOM cannot buy
ArduPilot board ID	7180 is free and below the ceiling; the request, the registry diff and the required <code>Readme.md</code> are ready in <code>firmware/</code> . The local build still uses 9001, which is above the ceiling the registry asks you to respect

18 Part 17 — What is not built yet

The board and its firmware are green. The project's *premise* is not, and a clean gate says nothing about it, so it is stated here rather than left to the repository's own notes.

The Iridium Doppler solver exists and is validated; the chain around it does not.

Piece	Status
The least-squares Doppler solve	built and validated — <code>tools/soop_solver.py</code> . Gauss-Newton over position and clock with an analytic Jacobian; it recovers a known position exactly from clean geometry, and reports <i>singular</i> on degenerate geometry rather than inventing a number
SGP4 and real TLE ephemeris	not built — the solver takes satellite state as <i>input</i> , and a circular propagator stands in for the test. That is enough to prove the geometry inverts; it is not an ephemeris
Burst detection and frequency estimation	not built — from I/Q, to the precision the solver needs. No RF bench run substitutes for writing it
The C firmware for the H743	not built
Getting the fix into the EKF	not built — an <code>AP_GPS</code> backend or <code>AP_ExternalAHRS</code> . <code>GPS_INPUT</code> is the documented route for an <i>external</i> computer, and this aircraft solves on board

The solver also established the receiver's requirement, which was written down nowhere before: **about 5 Hz of burst frequency precision over 40–80 bursts**. An 8.28 ms burst has roughly 120 Hz of raw FFT resolution, so the carrier must be estimated well inside a bin — that is the number T3b is really trying to meet.

19 The one-page order

1. `python3 tools/preflight.py` — must print **READY TO ORDER**
2. `bash tools/make_order_bundle.sh` — regenerate everything
3. Upload `fab/NAVCORE-SoOP-order-bundle.zip`, or the gerbers plus the default BOM/CPL pair
4. 6 layers, 45.1×46.1 mm, 1.6 mm, **1 oz** outer, ENIG, 4 mil, via 0.45/0.20

5. 5 bare boards, 2 assembled. Standard panelised 2×2 unless you want to hand-fit U3
6. Confirm Y1 = C2682774 and Y2 = C5563878; refresh the stock snapshot if it is over 7 days old
7. Order the aircraft parts, **including the M3 assortment** and the Tier 4 RF kit. Do not buy a PMW3901 or VL53L1X
8. While waiting: prepare the board ID request, print the padmap, decide the USB edge
9. On arrival: inspect under magnification, buzz the ESC cable, measure the tapped depths
10. T1 current-limited rails → T2 bootloader over SWD → T3 sensors, props off
11. T3a thermal (U8 needs a probe) → T3c IMU temp cal → T3d nine unlabelled pads
12. T3b before any position-hold claim: measure the real sigma
13. SITL scenarios → tethered hover → GPS Loiter. **Props on last**